

Mini Project

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Date : 22-08-2025

Question:

1. Create fastapi project along with sqllite to store username, password and role.
2. Create ASP.net c# project with MSSQL server as backend with two tables.
 - 2.1 one table to store the: ban account number, amount in account, created date.
 - 2.2 another table to store the customer details: name and age.
 - 2.3 each customer can have more than one accounts. link these tables via foreign key.
3. in c#, login api should call the fast api login api to check the username, password correctness.
4. once the credentials are ok, then the jwt token should be generated and provided.
5. only user with admin access can insert customer and back account details.
6. create an account balance api, which receives username as input and provides the balance amount accross all the accounts.
 - 6.1 logged in user can view the balance amount.

Solution:

Fast Api Code :

```
from sqlalchemy import create_engine, Column, Integer, String
from sqlalchemy.orm import sessionmaker, Session, declarative_base
from fastapi import FastAPI, Depends, HTTPException, status
from pydantic import BaseModel
from typing import List
import hashlib
import uvicorn
from fastapi.responses import RedirectResponse

url = "sqlite:///D:/python/Fastapi/user.db"
engine = create_engine(url, connect_args={"check_same_thread": False})
```

```
SessionLocal = sessionmaker(autocommit=False, autoflush=False,
bind=engine)

Base = declarative_base()

app = FastAPI()

class User(Base):
    __tablename__ = "user"
    Id = Column(Integer, primary_key=True, index=True)
    Name = Column(String(100), nullable=False)
    Password = Column(String(100), nullable=False)
    Role = Column(String(100), nullable=False)

Base.metadata.create_all(bind=engine)

class Users(BaseModel):
    Name: str
    Password: str
    Role: str

class UserResponse(BaseModel):
    Name: str
    Role: str

    class Config:
        from_attributes = True

class LoginRequest(BaseModel):
    name: str
    password: str

def get_db():
    db = SessionLocal()
    try:
        yield db
    finally:
        db.close()
```

```

def hash_password(password: str) -> str:
    return hashlib.sha256(password.encode('utf-8')).hexdigest()

def verify_password(stored_hash: str, password: str) -> bool:
    return stored_hash == hash_password(password)

@app.post("/add_user")
def add_user(user: Users, db: Session = Depends(get_db)):
    hashed_pw = hash_password(user.Password)
    u = User(Name=user.Name, Password=hashed_pw, Role=user.Role)
    db.add(u)
    db.commit()
    db.refresh(u)

    return RedirectResponse(url="/get_users", status_code=303)

@app.get("/get_users", response_model=List[UserResponse])
def get_users(db: Session = Depends(get_db)):
    return db.query(User).all()

@app.get("/get_userbyId/{id}", response_model=UserResponse)
def get_user(id: int, db: Session = Depends(get_db)):
    u = db.query(User).filter(User.Id == id).first()
    if not u:
        raise HTTPException(status_code=404, detail="User not found")
    return u

@app.post("/login", response_model=UserResponse)
def user_login(login_request: LoginRequest, db: Session =
Depends(get_db)):
    name = login_request.name
    password = login_request.password
    u = db.query(User).filter(User.Name == name).first()

```

```

        if u is None or not verify_password(u.Password, password):
            raise HTTPException(status_code=401, detail="Invalid username or password")

        return UserResponse(Name=u.Name, Role=u.Role)

if __name__ == "__main__":
    uvicorn.run("Fastapi:app", host="127.0.0.1", port=8000, reload=True)

```

Output:

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [24528]
INFO: Stopping reloader process [2844]
PS D:\python> python Fastapi.py
C:\Users\narendrakumar.s\AppData\Roaming\Python\Python312\site-packages\pydantic\_internal\_config.py:373: UserWarning: Valid config keys have changed in V2:
* 'orm_mode' has been renamed to 'from_attributes'
  warnings.warn(message, UserWarning)
INFO: Will watch for changes in these directories: ['D:\python']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [16704] using StatReload
C:\Users\narendrakumar.s\AppData\Roaming\Python\Python312\site-packages\pydantic\_internal\_config.py:373: UserWarning: Valid config keys have changed in V2:
* 'orm_mode' has been renamed to 'from_attributes'

```

Login User :

127.0.0.1:8000/docs#/default/add_user_add_user_post

```

-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "Name": "Naren",
  "Password": "admin",
  "Role": "Admin"
}'

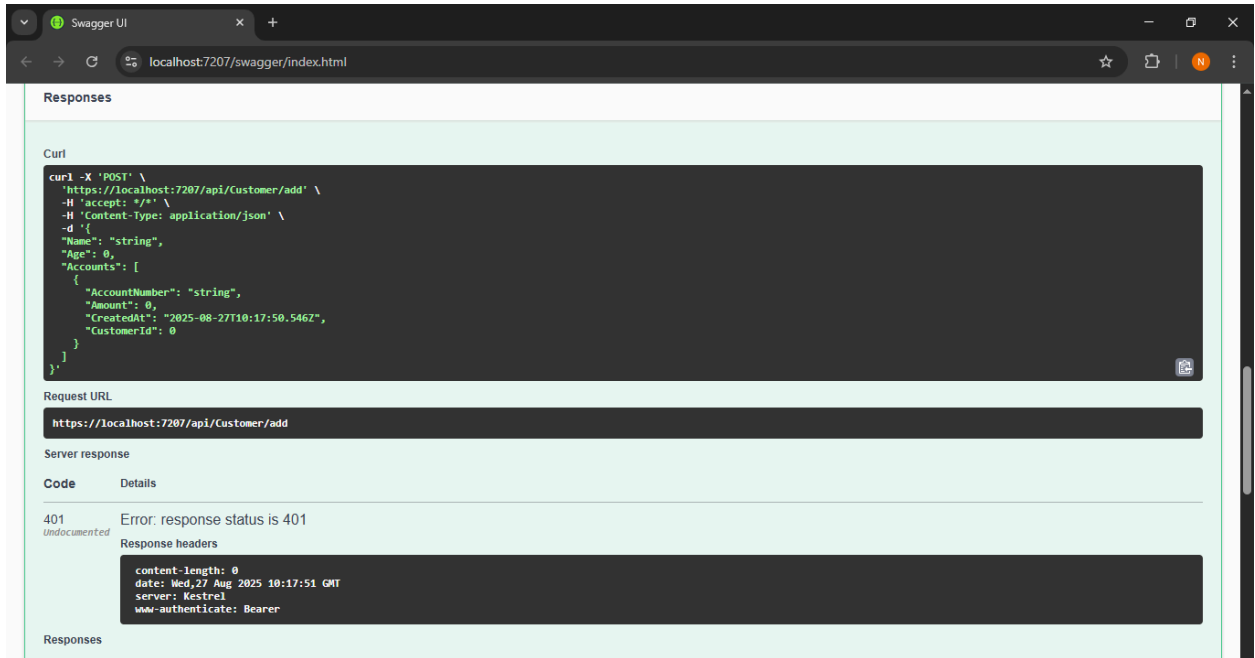
```

Request URL
http://127.0.0.1:8000/add_user

Server response

Code	Details
200	Response body <pre> [{ "Name": "Naren", "Role": "Admin" }, { "Name": "Praveen", "Role": "Admin" }, { "Name": "Jack", "Role": "Admin" }, { "Name": "Naren", "Role": "Admin" }] </pre> Response headers <pre> content-length: 130 content-type: application/json date: Wed, 27 Aug 2025 10:15:11 GMT server: uvicorn </pre>

Unauthorized Access:



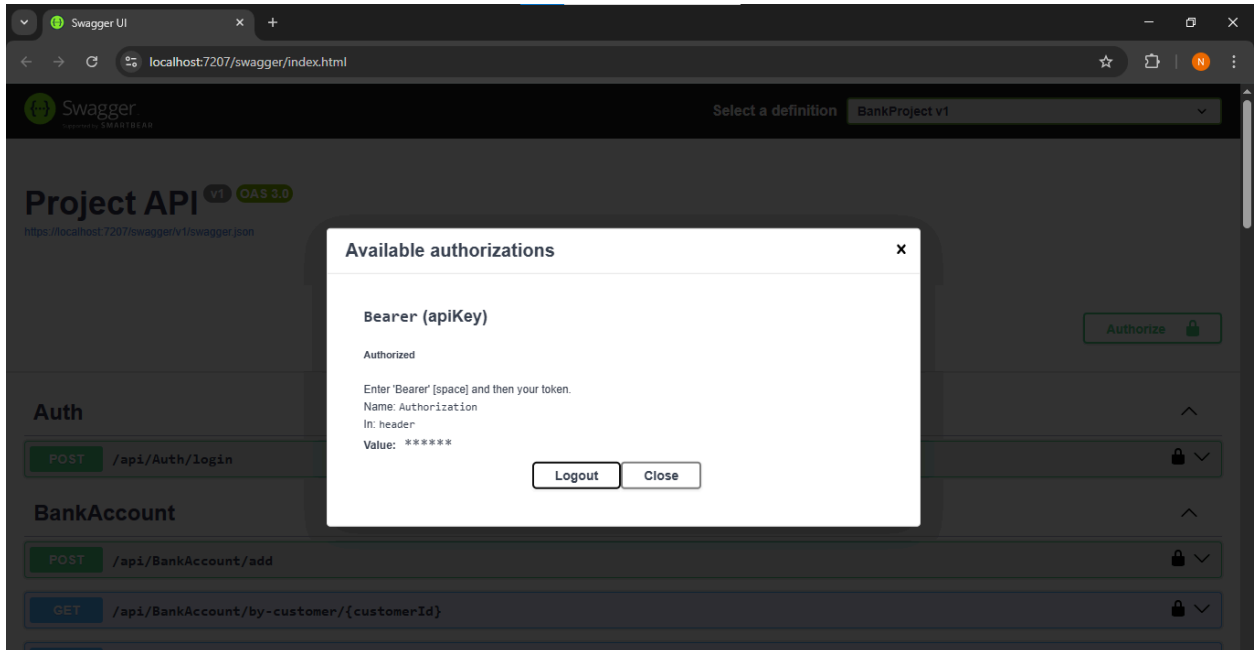
The image shows the Swagger UI interface in a web browser. The URL bar indicates the address is `localhost:7207/swagger/index.html`. The main content area displays the details of a failed request:

- Curly:** A cURL command is shown for a POST request to `https://localhost:7207/api/Customer/add` with a JSON body.
- Request URL:** `https://localhost:7207/api/Customer/add`
- Server response:** A table shows the response details.

Code	Details
401	Error: response status is 401
- Response headers:** A table lists the headers from the server response.

Header	Value
content-length	0
date	Wed, 27 Aug 2025 10:17:51 GMT
server	Kestrel
www-authenticate	Bearer

Authorization in C# :

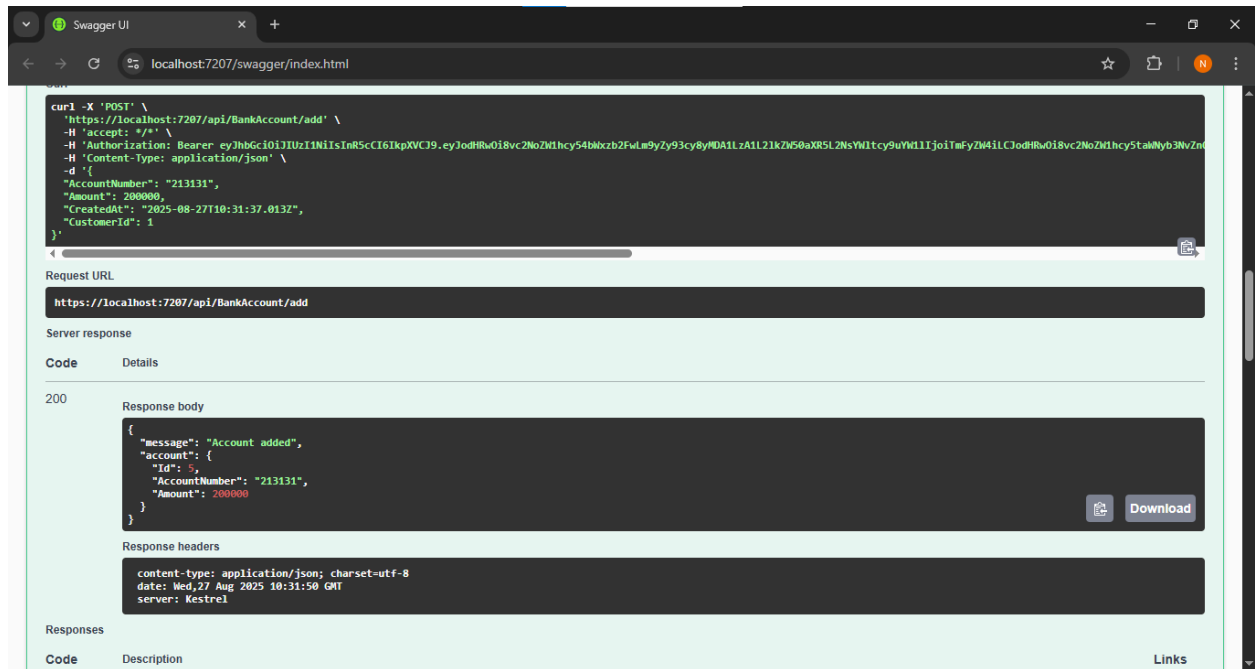


The image shows the Swagger UI interface with an 'Available authorizations' dialog box open. The dialog box is titled 'Available authorizations' and contains the following information:

- Bearer (apiKey)**
- Authorized**
- Enter 'Bearer' [space] and then your token.**
- Name:** Authorization
- In:** header
- Value:** *****

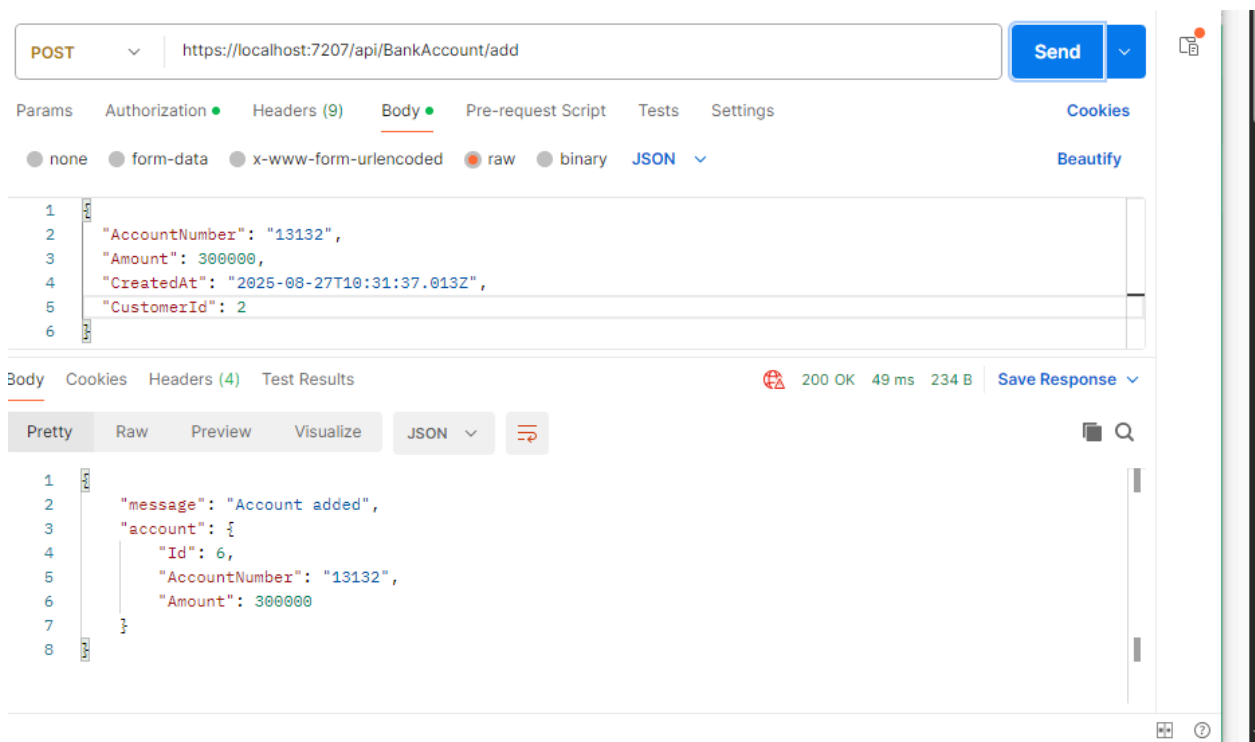
Buttons for 'Logout' and 'Close' are visible at the bottom of the dialog box. In the background, the Swagger UI shows the 'Project API v1 OAS 3.0' and a list of endpoints, including `POST /api/Auth/login`, `POST /api/BankAccount/add`, and `GET /api/BankAccount/by-customer/{customerId}`.

Adding the Bank Account in the Admin Login:



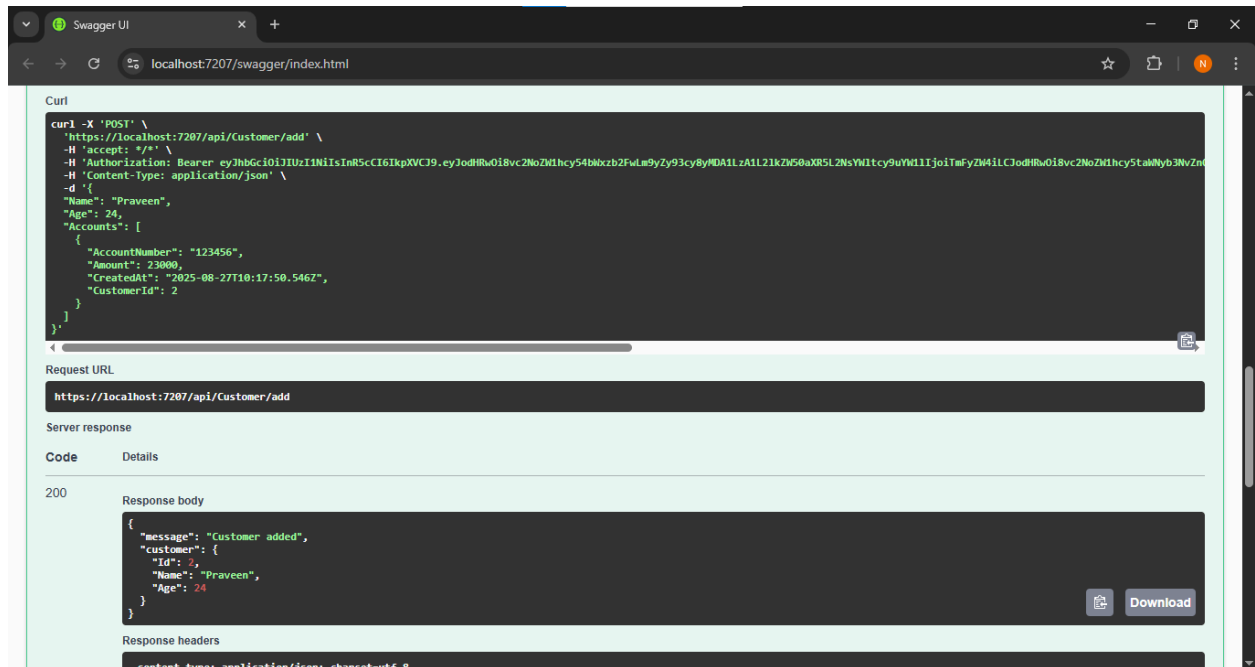
The image shows the Swagger UI interface in a web browser. The URL bar displays `localhost:7207/swagger/index.html`. The main area shows a cURL command for a POST request to `https://localhost:7207/api/BankAccount/add`. The request body is a JSON object: `{ "AccountNumber": "213131", "Amount": 200000, "CreatedAt": "2025-08-27T10:31:37.013Z", "CustomerId": 1 }`. The response is a 200 OK status with a JSON body: `{ "message": "Account added", "account": { "Id": 5, "AccountNumber": "213131", "Amount": 200000 } }`. The response headers include `content-type: application/json; charset=utf-8`, `date: Wed, 27 Aug 2025 10:31:50 GMT`, and `server: Kestrel`.

Adding the Bank Account in the Admin Login in Post Man :

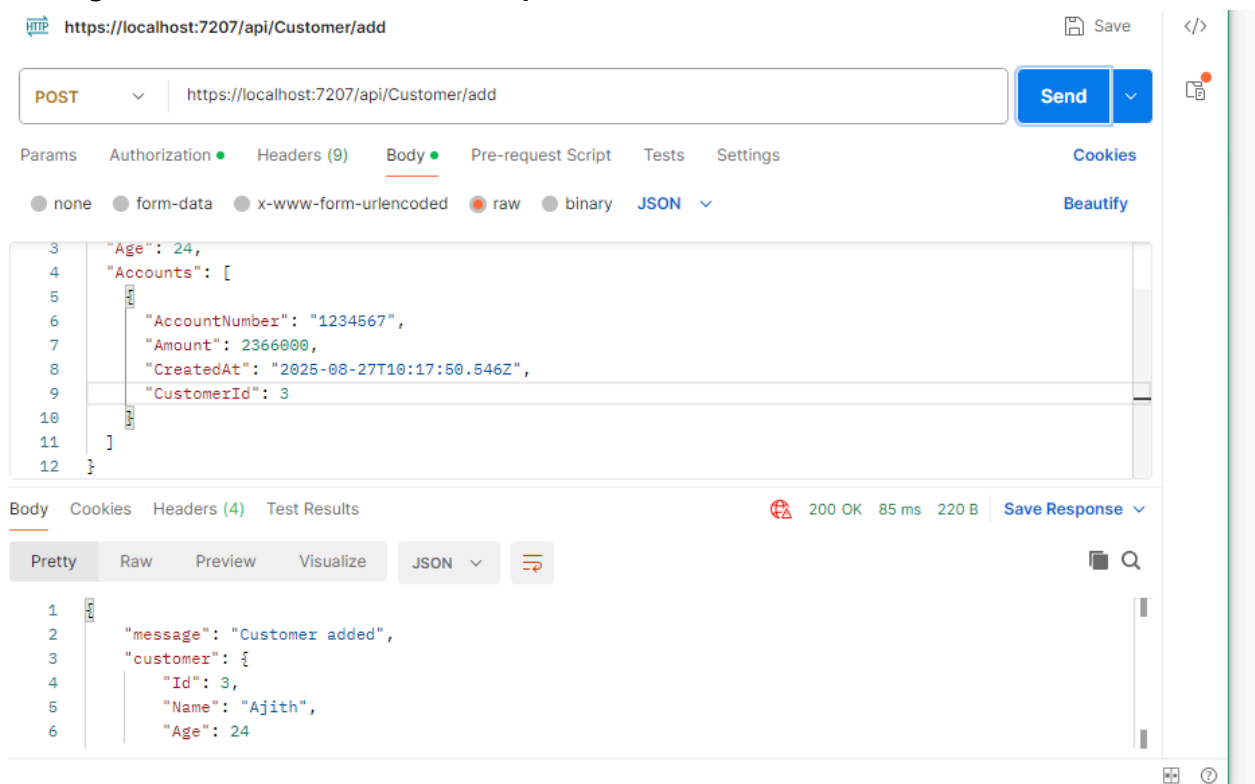


The image shows the Postman interface. The request method is `POST` and the URL is `https://localhost:7207/api/BankAccount/add`. The request body is a JSON object: `{ "AccountNumber": "13132", "Amount": 300000, "CreatedAt": "2025-08-27T10:31:37.013Z", "CustomerId": 2 }`. The response is a 200 OK status with a JSON body: `{ "message": "Account added", "account": { "Id": 6, "AccountNumber": "13132", "Amount": 300000 } }`. The response headers include `content-type: application/json; charset=utf-8`, `date: Wed, 27 Aug 2025 10:31:50 GMT`, and `server: Kestrel`.

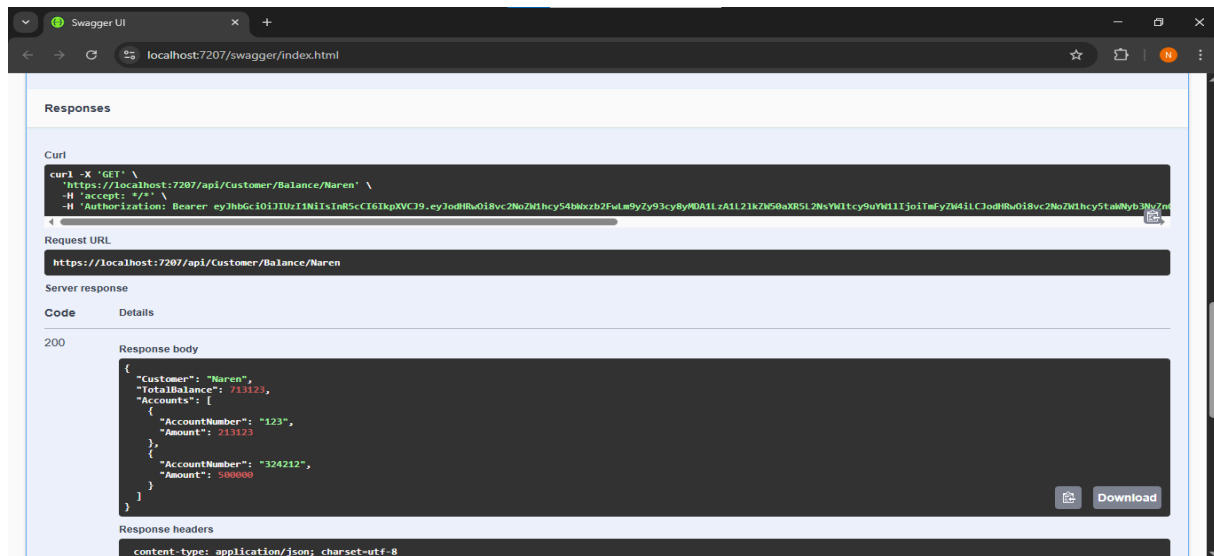
Adding the customer via Swagger Api :



Adding the Customer Via Post Man Api :



Getting Balance of the customer with customer name :



Getting Balance of the customer with customer name in Post Man :

